main(2).pdf
PDF

foundationally, adversarially and from first-principles attack this whitepaper I'm busy writing. the intent should be clear.

Attack the mathematics AND the physics. I'm both a

Show more

Thought for 9m 23s

I read the uploaded draft as a hostile referee would. The strongest defensible core is much narrower than the current framing: **you have a formal algebraic PBW-shadow theorem for a selected stable trace sector, plus a degree-zero Hamiltonian Poisson trace map. You do not yet have a closed-open string theorem, a cochain-level factorization-center map, or a deformation-level cotangent/descendant match.**

main

1. Main kill shot: the theorem is not “Koszul duality” in the advertised sense

Your theorem says the open stable Hamiltonian sector is

$$\mathcal{O}^{cl}_{\mathrm{brane},H} \cong S(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]),$$

while the closed observables are

$$\mathcal{O}^{cl}_{\mathrm{grav}} \cong C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}),$$

whose Koszul dual is

$$(\mathcal{O}^{cl}_{\mathrm{grav}})^{!} \simeq U(\mathfrak{g}).$$

But the actual comparison to the open side is only after passing to the **PBW special fibre**

$$U_{\hbar}(\mathfrak{g})/(\hbar) \cong S(\mathfrak{g}).$$

The paper itself states that the identification is “a filtered graded-commutative PBW special-fibre product identification, not a deformation-level identification of the descendant module actions.” That is a major downgrade, not a technical caveat. It means the open algebra is not shown to be the Koszul dual of the closed algebra; it is shown to match the **associated graded commutative shadow** of that Koszul dual, and only in a selected sector. main

So the theorem title “Local stable Hamiltonian-sector Koszul duality” is too strong. A more honest title would be:

> **PBW special-fibre comparison for the stable Hamiltonian trace sector.**

If you keep the current title, a referee can fairly say the title claims a deformation-level Koszul duality theorem that the body explicitly disclaims.

2. The closed-open map is not a map from closed observables

This is the deepest mathematical obstruction, and your own draft proves it.

You define closed local observables as CE cochains

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}).$$

But your boundary assignment

$$f \mapsto \mathrm{Tr}\, f(\phi_1, \phi_2)$$

starts from a **Hamiltonian mode** $f \in \bar{A}$, not from a closed CE cochain. Proposition 0.2.18 proves that $\bar{A}$ is perfect and that the CE length-one coordinate dual to a Hamiltonian mode is not closed; no CE cocycle has that length-one part. The draft explicitly concludes that the boundary assignment is “a Hamiltonian-mode, or PBW/Koszul-dual, construction” and “not a cochain-level map out of $C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g})$.” main

That means the physical closed-open slogan is currently wrong at the observable level. A closed-open map should be something like

$$\mathrm{Obs}^{cl}_{\mathrm{closed}\ \partial} \longrightarrow Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{cl}_{\mathrm{open}}),$$

but you do not construct that. You construct a map from Hamiltonian generators / PBW modes to boundary functions.

This is not a cosmetic source-convention issue. It is a categorical-level mismatch:

$$\text{closed observable} \neq \text{Hamiltonian Lie algebra element}.$$

Your Problem 0.2.20 states exactly this, but then the theorem language keeps sounding as if the closed-open map exists. It does not.

3. The cotangent/descendant sector does not match the closed BF module

The closed Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1],$$

so the $\mathfrak{h}^\vee[1]$ part must carry the **coadjoint** action of $\mathfrak{h}$. But the polynomial one-ψ descendants carry the wrong action. Your Corollary 0.2.45 proves this explicitly:

$$F_{p,q} : \Psi_{a,b} \mapsto (pb - qa)\Psi_{a+p-1,b+q-1},$$

whereas the closed coadjoint action is

$$\mathrm{ad}^\vee_{z_1^p z_2^q}\, \delta_{a,b} = -(pb - qa + p - q)\delta_{a-p+1,b-q+1}.$$

Even the direction is wrong in the simplest case: z_1 lowers the open descendant index b, while it raises the closed coadjoint dual index. The draft itself says the polynomial descendants do not realize the coadjoint deformation-level action. main

This is devastating for the advertised closed cotangent BF comparison. The best you currently have is:

$$\mathfrak{h}^\vee[1] \cong \langle \Psi_{a,b} \rangle$$

as a **graded vector space / PBW special-fibre label set**, not as an $\mathfrak{h}$-module, not as a Lie algebra representation, and not as a deformation-compatible boundary sector.

Your principal-part repair is mathematically natural, but it is not part of the open polynomial brane theory. The paper admits it must be promoted to a factorization object before it can participate in the full cotangent lift. main

So the descendant part of the theorem should be phrased as a **basis-level identification only**. Anything stronger is false.

4. Corollary 0.2.51 is false as written unless you explicitly quotient out constants

Take the simplest possible test:

$$f = z_1, \qquad g = z_2.$$

In the reduced Hamiltonian algebra

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1,$$

one has

$$\{z_1, z_2\}_{\bar{A}} = 1 \mod \mathbb{C} = 0.$$

So your Corollary 0.2.51 predicts

$$\{I_\partial(a, z_1), I_\partial(b, z_2)\} = 0.$$

But on the open matrix side,

$$I_\partial(a, z_1) = \int a(t) \operatorname{Tr} \phi_1(t)\, dt, \qquad I_\partial(b, z_2) = \int b(t) \operatorname{Tr} \phi_2(t)\, dt.$$

Using

$$\{(\phi_1)^i_j, (\phi_2)^k_l\} = \delta^i_l \delta^k_j,$$

one gets

$$\{\operatorname{Tr} \phi_1, \operatorname{Tr} \phi_2\} = N.$$

Therefore

$$\{I_\partial(a, z_1), I_\partial(b, z_2)\} = N \int a(t) b(t)\, dt,$$

not zero.

Theorem 0.2.50 is safe because it explicitly works in the connected Hamiltonian trace quotient where constants vanish. But Corollary 0.2.51 states the smeared identity without repeating that quotient, even though the displayed formula fails for the most basic generators. main

This is not a minor edge case. It is the central extension of the trace Poisson algebra. Physically, it is the center-of-mass symplectic pairing of the brane stack. You cannot silently throw it away in a “bulk-boundary coupling” statement.

Fix: every bracket identity involving z_1, z_2 must say

after projection to the Hamiltonian quotient by scalar trace classes.

Without that, the formula is wrong.

5. The “central operation” theorem is too close to tautological

Theorem 0.2.17 constructs an interval-wise operation by multiplication with

$$\int_I a(t)\,\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))\, dt.$$

But in the reduced open model the observable algebra is graded-commutative, so multiplication by any closed element is automatically an ordinary Hochschild central operation. The paper itself says the proof uses $Q\phi_i = 0$, graded-commutativity, and then the Poisson identity after projection. main

So the ordinary E_1-centrality part is not evidence of a closed-open theorem. The only nontrivial part is the Poisson bracket identity, and that identity is degree-zero, quotient-dependent, and not a full P_0 factorization-center map.

This should be demoted rhetorically. It is a useful check, but not a central construction of closed-open string theory.

6. The “sector” is selected by hand

The open Hamiltonian algebra is defined as the sector generated by:

1. nonempty primitive degree-zero traces;
2. nonconstant primitive one-ψ Koszul classes.

Then you exclude:

1. the scalar trace $\mathrm{Tr}(1) = N$;
2. $\mathrm{Tr}(\psi)$;
3. all higher-ψ primitive homology;
4. finite-N trace identities;
5. the genuine coadjoint action mismatch.

The draft is honest that “sector” is literal and that higher-ψ homology lies outside the sector. main

But then the introduction and theorem language still make it sound like the open brane algebra itself is being matched to the closed Koszul dual. It is not. You are matching a deliberately carved-out subquotient/subsector.

A hostile formulation would be:

> The theorem proves that, after discarding the scalar central extension, discarding higher antifield classes, discarding the scalar antifield, stabilizing away finite-N identities, forgetting the closed coadjoint module action, and passing to the PBW associated graded, the remaining symbols can be placed in bijection.

That may sound brutal, but it is structurally accurate.

7. The large-N limit is algebraic, not physical

Your stable large-N limit is degreewise invariant theory under block restriction. That is a valid algebraic stabilization. But it is not the physical large-N limit used in matrix models unless you also specify:

$$\mathrm{Tr} \quad \text{versus} \quad \frac{1}{N}\,\mathrm{Tr},$$

connected correlators/cumulants, scaling of $\hbar$, planar versus nonplanar terms, and the fate of central terms like

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N.$$

Right now "connected stable trace sector" is doing several jobs at once: removing the empty trace, selecting primitive single traces, killing constants, and evading finite-rank trace identities. Those are not the same operation physically.

You need a separate definition:

$$\mathrm{Prim}_{\mathrm{st}}, \qquad \mathrm{Conn}_{N\to\infty}, \qquad \text{Hamiltonian scalar quotient},$$

and you should not conflate them.

8. The reduced gauge/BV model suppresses a real quotient-stack issue

You say you take GL_N-invariants in characteristic zero, so the Chevalley part of gauge BRST acts trivially on trace functions. That is fine for trace functions, but it does not by itself identify the full BV quotient with the invariant Koszul complex.

There are two different models hiding here:

1. **Algebraic quotient by a linearly reductive group**, where taking invariants is exact.

2. **Lie-algebra/BRST Chevalley-Eilenberg quotient**, where gauge ghosts can have their own cohomology unless a precise presentation kills them.

Your Remark 0.2.67 tries to justify this, but the main calculation uses the reduced invariant complex as if the quotient-stack subtleties were automatically absent. That is acceptable only if you explicitly state:

> We are computing the algebraic invariant functions on the derived commuting scheme, not the full CE/BRST cohomology of the infinitesimal gauge algebra.

Otherwise a referee can object that "the Chevalley part acts trivially on trace functions" is not the same as "the Chevalley sector has disappeared."

9. The one-ψ homology proof is plausible but under-proved

Proposition 0.2.28 is one of the technically interesting parts of the paper. But the proof leans heavily on the chip-moving state complex and an Abel-map argument. The draft says the quotient state complex has the homology of S^1, with convex fibres and affine stabilizer quotients. main

That is the kind of argument that will get attacked unless made completely formal. The danger points are:

1. the passage from cyclic words with graded signs to cellular chains;
2. the quotient by cyclic relabelling;
3. stabilizers of configurations with symmetry;
4. compatibility of orientations/signs with the two-ψ boundary;
5. the claim that higher cells do not affect H_1;
6. the identification of the particular generator with $\Psi_{k,l}$.

The computational script checks up to small degree, but that does not replace the proof. main

If this proposition is essential, give it a standalone, fully combinatorial proof, or relegate it to a conjectural/computational lemma. Right now it is probably the most vulnerable purely algebraic proof in the paper.

10. The Tate/Koszul-duality formalism is under-specified

The claim

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g})^{!} \simeq U(\mathfrak{g})$$

for a completed Tate Lie algebra is believable, but the draft's Lemma 0.2.40 is doing a huge amount of foundational work. A hostile reader will ask:

1. What exact category of complete filtered Tate spaces is being used?
2. What are the admissible quasi-isomorphisms?
3. Is the CE coalgebra genuinely conilpotent after coefficient completion?
4. Which completed bar/cobar adjunction is being invoked?
5. Are the dualizations continuous in both length and Taylor degree?
6. Is the PBW filtration complete and separated?
7. Does the coadjoint action preserve the claimed topology for arbitrary formal Hamiltonians?

You gesture at these, but not enough to make the theorem mechanically checkable. Since this lemma underwrites the closed-side Koszul dual, it cannot remain a sketch.

11. The closed sector is a formal BF model, not yet a physical closed string sector

The paper distinguishes itself from Gwilliam–Williams and says it is not their holomorphic bosonic string. Good. But it still uses phrases like "mixed holomorphic-topological strings," "twisted M-theory," "closed string sector," and "brane coupling." The actual constructed object is a formal local Hamiltonian cotangent BF theory on

$$\mathbb{R}^2 \times \mathbb{C}^2$$

with coefficient Lie algebra

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}.$$

That is a perfectly reasonable BV toy model. But the physics origin is not derived. In particular:

1. No parent 11-dimensional M-theory background is specified.
2. The dimension count is not explained.
3. The twist is not constructed.
4. The brane boundary condition/defect theory is not derived from a worldvolume theory.
5. No anomaly cancellation/QME theorem is proved.
6. No propagator or defect kernel is constructed.
7. The quantum graph realization is explicitly left open.

So the physics claim should be downgraded from "this is a local mixed string model" to:

> This is a formal local BV model motivated by the Hamiltonian sector expected in a mixed holomorphic-topological brane setup.

That wording is much harder to attack.

12. The BCOV/Kodaira-Spencer analogy is dangerous

You describe the closed sector as Hamiltonian Kodaira-Spencer/BF. But the actual construction is not BCOV theory on a Calabi-Yau threefold, and not a direct truncation of BCOV. You use divergence-free vector fields on a holomorphic symplectic surface, identify them locally with Hamiltonians modulo constants, and then add a cotangent BF partner.

That is mathematically coherent, but the phrase “Kodaira-Spencer” invites the wrong comparison. In BCOV, the field complex, divergence operator, anomaly structure, and quantization problem are much more rigid. Here the Hamiltonian reduction is a chosen local sector.

I would use “Hamiltonian BF theory for the formal symplectic disk” as the primary name, and mention Kodaira-Spencer only as motivation.

13. The quantum section is a target, not a result

The Weyl/Moyal formulas are correct as algebraic deformation-quantization targets for the constant Poisson disk. But the draft is clear that the actual quantum boundary theorem is not proved. Proposition 0.2.52 only proves BRST descent of individual Weyl-ordered traces; the Moyal commutator identity in quantum Hamiltonian reduction is left as Problem 0.2.54. The draft also identifies the normal-ordering/Capelli shift

$$YX - XY + \hbar N I = 0$$

as an obstruction that can feed lower-degree traces into the connected sector. main

This is not a small technical issue. It is exactly where the classical trace-bracket story can fail quantum mechanically.

So any sentence suggesting that the open quantum brane algebra **is** the Weyl/Moyal algebra should be changed to:

> The Weyl/Moyal algebra is the proposed formal target; the reduced finite-N and stable connected identities remain open.

Your draft mostly says this, but the abstract and introduction should be ruthlessly consistent.

14. The central extension is physically meaningful, not disposable bookkeeping

Classically,

$$\{z_1, z_2\} = 1$$

becomes zero in $\bar{A}$, because constants are removed. But on the brane,

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N.$$

You remove this by projecting out scalar trace classes. Algebraically that is allowed for the reduced Hamiltonian vector-field algebra. Physically it is not innocent: it removes the center-of-mass canonical pair of the brane stack.

This same issue reappears quantum mechanically as the normal-ordering shift. So the central term is not just a nuisance; it is the obstruction that keeps returning.

I would make this a named obstruction:

> **Scalar Hamiltonian / center-of-mass anomaly.**

Then every theorem should state whether it is before or after killing it.

15. The figures weaken the paper

The schematic ovals on pages 41, 50, and 51 do not communicate mathematical data: no labels, no propagator convention, no orientation, no automorphism factor, no boundary ordering, no kernel. The captions say they are not Feynman-integral proofs. That makes them look ornamental.

Either remove them or replace them with real graph notation:

$$\Gamma_1, \quad \Gamma_3, \quad \mathrm{Aut}(\Gamma), \quad P_\partial, \quad \text{ordered boundary inputs.}$$

As drawn, they invite the accusation that the graph discussion is gestural.

16. Reference and notation cleanup matters

There are malformed citation labels in the body, for example things like “[13]razmyslov” and “[2]tsygan.” The PBW discussion appears to cite “[16]loday-vallette,” but [16] is

Priddy and Loday–Vallette is [17]. These are small, but in a paper already making delicate claims, citation sloppiness reduces trust.

17. What the paper can safely claim

A defensible theorem would be something like:

> Let
>
> $$R_N = \mathbb{C}[\phi_1, \phi_2] \otimes \Lambda(\psi), \qquad Q\psi = [\phi_1, \phi_2],$$
>
> with GL_N-invariants and degreewise stable trace limit. In the selected Hamiltonian sector generated by nonempty degree-zero primitive traces and nonconstant primitive one-ψ classes, the stable open algebra is
>
> $$S(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]).$$
>
> The closed formal Hamiltonian BF Lie algebra is
>
> $$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{cont}[1].$$
>
> The PBW associated graded of $U(\mathfrak{g})$ has degreewise polynomial part
>
> $$S(\bar{A} \oplus \mathfrak{h}^{\vee}[1]).$$
>
> After choosing the monomial-dual basis, this graded-commutative algebra is identified with the selected stable open sector. On degree-zero Hamiltonians, the boundary evaluation
>
> $$f \mapsto \mathrm{Tr}\, f(\phi_1, \phi_2)$$
>
> is a Lie algebra map after quotienting scalar trace classes. The polynomial one-ψ descendants do not realize the closed coadjoint action; the full CE/factorization-center and cotangent boundary maps remain open.

That version is strong, precise, and hard to refute.

Bottom line

The algebraic skeleton is interesting and mostly coherent. The vulnerable claims are the words around it: “string,” “closed-open,” “Koszul duality,” “cotangent BF sector,” “factorization center,” and “quantum Moyal upgrade.” Right now those phrases outrun the proved content.

The paper becomes much stronger if you openly demote the main result to a **PBW special-fibre Hamiltonian trace comparison** and isolate three unsolved obstructions as first-class problems:

$$\text{CE-source obstruction}, \qquad \text{coadjoint descendant mismatch}, \qquad \text{scalar/Capelli cer}$$

Those are not peripheral. They are the actual mathematical and physical boundary of the current result.

Sources

Sources

No you don't seem to get it.

The point is NOT to cut the paper down to size. it's to understand what it is trying to prove and produce those constructions, derivations, deductions & mathematics and physics
Let's construct, deduct, prove the open closed derived center structure which is the central unresolved issue.

Show less

Thought for 13m 43s

You are right. The way to solve the central obstruction is not to shrink the claim; it is to identify the **correct derived-center model**. The missing construction is this:

$$C^{\bullet}_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^{\vee}[1]) \;\cong\; Z^{P_0}_{\mathrm{fact}}(S(\mathfrak{h}))$$

where the right side is not the ordinary commutative center. It is the **derived Poisson/factorization center**, modeled by Poisson cochains of the open Hamiltonian algebra. Under this identification, the cotangent CE generators map to the actual boundary Hamiltonians

$$u_f \longmapsto O_f = \mathrm{Tr}\, f(\phi_1, \phi_2),$$

while the Hamiltonian CE ghost generators map to the corresponding constant polyvector fields. This is exactly the cochain-level lift that the draft currently lists as unresolved in Problem 0.2.20. The point is that the naive source obstruction disappears because the boundary observable is **not** the image of the CE coordinate dual to $f \in \mathfrak{h}$; it is the image of the CE coordinate dual to the shifted cotangent generator $f^\vee[1]$. That is the derived-center mechanism. main

Below is the construction I would insert as a new section.

Derived open-closed center theorem

Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$$

with reduced Poisson bracket

$$[f, g] = \{f, g\}_{\bar{A}}.$$

Let

$$\mathfrak{g} = T^*[1]\mathfrak{h} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1],$$

with bracket

$$[f, g] = \{f, g\}, \qquad [f, \eta_\lambda] = \eta_{\mathrm{ad}^\vee_f \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0.$$

Let the stable degree-zero open Hamiltonian algebra be

$$A_\partial = S(\mathfrak{h}),$$

with generator O_f corresponding to

$$O_f = \mathrm{Tr}\, f(\phi_1, \phi_2).$$

Equip A_∂ with the Kirillov-Poisson bracket

$$\{O_f, O_g\}_\partial = O_{\{f,g\}}.$$

Then the derived P_0-factorization center of the locally constant open algebra with stalk A_∂ is modeled by the Poisson cochain complex

$$Z^{P_0}_{\mathrm{fact}}(A_\partial) \simeq \mathrm{PV}(A_\partial), \qquad d_{\mathrm{center}} = [\pi_\partial, -],$$

where π_∂ is the linear Poisson bivector determined by the bracket on $\mathfrak{h}$.

The claim is:

$$C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]\big) \cong (\mathrm{PV}(S(\mathfrak{h})), [\pi_\partial, -])\,.$$

This is the closed-open derived center map. In fact, in the formal stable Hamiltonian sector it is an isomorphism.

Explicit coordinates

Choose a homogeneous topological basis

$$H_I = z_1^{a_I} z_2^{b_I}, \qquad a_I + b_I > 0,$$

of $\mathfrak{h}$. Write

$$[H_I, H_J] = C_{IJ}^K H_K.$$

Concretely,

$$[H_{p,q}, H_{r,s}] = (ps - qr) H_{p+r-1,q+s-1},$$

with the convention that constant targets vanish.

Let O_I be the corresponding open boundary Hamiltonian

$$O_I = \mathrm{Tr}\, H_I(\phi_1, \phi_2).$$

The open Poisson bracket is

$$\{O_I, O_J\}_\partial = C_{IJ}^K O_K.$$

Therefore the Poisson bivector on the formal affine space $\mathfrak{h}^\vee$ is

$$\pi_\partial = \frac{1}{2} C_{IJ}^K O_K \frac{\partial}{\partial O_I} \wedge \frac{\partial}{\partial O_J}.$$

Let

$$\theta^I = \frac{\partial}{\partial O_I}$$

be the constant polyvector field dual to O_I. Then

$$\mathrm{PV}(A_\partial) = S(O_I) \otimes \Lambda(\theta^I).$$

The Poisson differential is

$$d_\pi = [\pi_\partial, -].$$

It acts by

$$d_\pi \theta^K = -\frac{1}{2} C^K_{IJ} \theta^I \theta^J,$$

and

$$d_\pi O_K = C^L_{KJ} O_L \theta^J.$$

Those two formulas are the entire derived-center mechanism.

Closed CE algebra of the cotangent BF sector

The closed Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

Let c^I be the CE coordinate dual to the Hamiltonian generator $H_I \in \mathfrak{h}$. Let u_I be the CE coordinate dual to the shifted cotangent generator $\eta_{\delta_I} \in \mathfrak{h}^\vee[1]$.

Thus

$$C^\bullet_{\mathrm{CE}}(\mathfrak{g}) = S(u_I) \otimes \Lambda(c^I),$$

up to the usual cohomological shifts.

The CE differential is

$$d_{\mathrm{CE}} c^K = -\frac{1}{2} C^K_{IJ} c^I c^J,$$

and

$$d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J.$$

This is the standard CE differential for the semidirect product

$$\mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

Now define

$$\Phi(c^I) = \theta^I, \qquad \Phi(u_I) = O_I.$$

Then

$$d_\pi \Phi(c^K) = d_\pi \theta^K = -\frac{1}{2} C^K_{IJ} \theta^I \theta^J = \Phi(d_{\mathrm{CE}} c^K),$$

and

$$d_\pi \Phi(u_K) = d_\pi O_K = C^L_{KJ} O_L \theta^J = \Phi(d_{\mathrm{CE}} u_K).$$

Therefore

$$\Phi d_{\mathrm{CE}} = d_\pi \Phi.$$

Since both source and target are freely generated completed graded-commutative algebras on the same generators, Φ is an isomorphism of complexes.

Moreover, the canonical shifted cotangent bracket on

$$C^\bullet_{\mathrm{CE}}\big(T^*[1]\mathfrak{h}\big)$$

satisfies

$$\{c^I, u_J\} = \delta^I_J, \qquad \{c^I, c^J\} = 0, \qquad \{u_I, u_J\} = 0.$$

The Schouten bracket on polyvectors satisfies

$$[\theta^I, O_J] = \delta^I_J, \qquad [\theta^I, \theta^J] = 0, \qquad [O_I, O_J] = 0$$

as a bracket on the underlying polyvector algebra. Thus Φ also preserves the P_0 bracket.

So the cochain-level closed-open center theorem is:

$$\Phi : C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]\big) \quad \sim \quad Z^{P_0}_{\mathrm{fact}}(S(\mathfrak{h}))$$

with

$$c^I \mapsto \frac{\partial}{\partial O_I}, \qquad u_I \mapsto O_I.$$

This is the missing construction.

Why this resolves the CE-source obstruction

The obstruction in the draft says that the CE coordinate c^I dual to $H_I \in \mathfrak{h}$ is not closed because $\mathfrak{h}$ is perfect. That is correct.

But the boundary Hamiltonian

$$O_I = \operatorname{Tr} H_I(\phi_1, \phi_2)$$

should not be identified with c^I. It should be identified with u_I, the coordinate dual to the shifted cotangent generator $\eta_{\delta_I} \in \mathfrak{h}^\vee[1]$.

The differential of u_I is not zero. It is

$$d_{\mathrm{CE}} u_I = C^L_{IJ} u_L c^J.$$

On the open side, this becomes

$$d_\pi O_I = [\pi_\partial, O_I] = C^L_{IJ} O_L \theta^J.$$

That is exactly the Hamiltonian vector field of the boundary observable O_I.

So the boundary Hamiltonian is not a closed scalar element of the derived center. It is a **degree-zero function inside the Poisson cochain algebra**, whose differential is its Hamiltonian vector field. This is precisely what a derived P_0-center should do.

In words:

$$\text{closed Hamiltonian ghost} \longmapsto \text{constant polyvector field},$$

while

$$\text{closed cotangent coordinate} \longmapsto \text{boundary Hamiltonian function}.$$

This is the structural point missing from the draft.

Recovery of the boundary bracket theorem

The ordinary Hamiltonian boundary map is the moment-map shadow of Φ:

$$J : \mathfrak{h} \longrightarrow A_\partial, \qquad J(f) = O_f = \operatorname{Tr} f(\phi_1, \phi_2).$$

The open Poisson bracket gives

$$\{J(f), J(g)\}_\partial = J(\{f, g\}).$$

For monomials,

$$f = z_1^k z_2^l, \qquad g = z_1^m z_2^n,$$

this becomes

$$\{\mathrm{Tr}(\phi_1^k \phi_2^l), \mathrm{Tr}(\phi_1^m \phi_2^n)\} = (kn - lm)\,\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1}),$$

with constant targets projected out.

That is the degree-zero current theorem in the paper, but now it is not isolated. It is the degree-zero shadow of the derived-center isomorphism

$$C^\bullet_{\mathrm{CE}}(T^*[1]\mathfrak{h}) \cong \mathrm{PV}(S(\mathfrak{h})).$$

Factorization version on intervals

For an interval $I \subset \mathbb{R}$, define the current algebra

$$\mathfrak{h}_I = C_c^\infty(I) \otimes \mathfrak{h}$$

with bracket

$$[a \otimes f, b \otimes g] = ab \otimes \{f, g\}.$$

Define the open Hamiltonian factorization algebra by

$$A_\partial(I) = S(\mathfrak{h}_I),$$

with extension by zero for inclusions of intervals and ordinary multiplication for disjoint intervals.

The Poisson bracket is

$$\{O_f(a), O_g(b)\} = O_{\{f,g\}}(ab),$$

where

$$O_f(a) = \int_I a(t)\,\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))\, dt.$$

The derived P_0-center is

$$Z^{P_0}_{\mathrm{fact}}(A_\partial)(I) \simeq \mathrm{PV}(A_\partial(I)), \qquad d = [\pi_I, -].$$

The closed restricted cotangent BF observable algebra is

$$C^\bullet_{\mathrm{CE}}\left(\mathfrak{h}_I \ltimes \mathfrak{h}_I^\vee[1]\right).$$

The interval-wise map is

$$c_a^I \longmapsto \theta_a^I, \qquad u_{I,a} \longmapsto O_I(a),$$

where θ_a^I is the constant polyvector field dual to the current coordinate $O_I(a)$.

The same coordinate calculation proves

$$C^\bullet_{\mathrm{CE}}\left(\mathfrak{h}_I \ltimes \mathfrak{h}_I^\vee[1]\right) \cong Z^{P_0}_{\mathrm{fact}}(A_\partial)(I).$$

The construction is compatible with extension by zero and factorization over disjoint intervals because all brackets are pointwise-current brackets and the test functions multiply as ab. This upgrades the stalkwise theorem to the locally constant factorization statement.

Where the one-ψ descendants enter

The derived-center theorem above uses

$$A_\partial = S(\mathfrak{h})$$

as the open Hamiltonian algebra. Its derived center already contains the cotangent BF data because

$$Z^{P_0}(S(\mathfrak{h})) \simeq \mathrm{PV}(S(\mathfrak{h})) \simeq C^\bullet_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]).$$

So the cotangent direction of the closed theory does **not** have to be represented as an extra ordinary polynomial open field. It appears as the polyvector/derived-center direction.

The one-ψ descendants instead belong to the **Koszul-dual/PBW open algebra**

$$S(\mathfrak{h} \oplus \mathfrak{h}^\vee[1]).$$

That is the PBW special fibre of

$$U(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]).$$

The correct relation is therefore:

$$\text{closed CE observables} \;\cong\; \text{derived } P_0 \text{ center of } S(\mathfrak{h}),$$

while

$$\text{Koszul dual of closed CE observables} \;\rightsquigarrow\; \text{PBW open trace sector } S(\mathfrak{h} \oplus \mathfrak{h}^\vee[1]).$$

The polynomial $\Psi_{a,b}$ classes are valid PBW-special-fibre representatives, but they should not be forced to realize the coadjoint action inside the derived center. If one wants an honest deformation-level realization of the $\mathfrak{h}^\vee[1]$ PBW generators, then one should use the principal-part module

$$P = \bigoplus_{a,b\geq 0,\; a+b>0} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

with action

$$f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\}.$$

Then

$$\mathfrak{h}^\vee_{\mathrm{cont}} \cong P$$

as an $\mathfrak{h}$-module, and the PBW-special-fibre open algebra becomes

$$S(\mathfrak{h} \oplus P[1]).$$

In that enlarged PBW model,

$$J(f) = O_f, \qquad J(\rho[1]) = \Theta_\rho,$$

with brackets

$$\{O_f, O_g\} = O_{\{f,g\}},$$

$$\{O_f, \Theta_\rho\} = \Theta_{f\cdot\rho},$$

$$\{\Theta_\rho, \Theta_\sigma\} = 0.$$

This gives the deformation-correct descendant action.

The physical interpretation

The closed Hamiltonian BF sector is the shifted cotangent theory

$$T^*[1]B\mathfrak{h}.$$

Its functions are

$$C^\bullet_{\mathrm{CE}}(T^*[1]\mathfrak{h}) = C^\bullet_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]).$$

The open brane degree-zero phase space is the large-N Hamiltonian trace phase space with coordinate algebra

$$S(\mathfrak{h}).$$

The derived Poisson center of this open phase space is the algebra of polyvector fields

$$\mathrm{PV}(S(\mathfrak{h})).$$

For a linear Poisson structure, the Poisson cochain complex is exactly CE cochains of the shifted cotangent Lie algebra:

$$\mathrm{PV}(S(\mathfrak{h})) \cong C^\bullet_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]).$$

So physically:

$$\text{closed BF fields} = \text{polyvector fields on the open Hamiltonian trace phase space.}$$

The boundary evaluation

$$f(z_1, z_2) \longmapsto \mathrm{Tr}\, f(\phi_1, \phi_2)$$

is the moment map coordinate inside this polyvector algebra. Its Poisson differential is the Hamiltonian vector field it generates. That is why it was wrong to demand that it be a closed CE length-one class.

Replacement theorem for the paper

I would replace Problem 0.2.20 by the following theorem.

Theorem. Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$$

with reduced Poisson bracket, and let

$$A_\partial = S(\mathfrak{h})$$

be the stable connected degree-zero Hamiltonian trace algebra with Poisson bracket

$$\{O_f, O_g\} = O_{\{f,g\}}.$$

Then the closed Hamiltonian cotangent BF local observable algebra satisfies

$$\mathrm{Obs}^{cl}_{\mathrm{closed},H} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \cong Z^{P_0}_{\mathrm{fact}}(A_\partial).$$

Under this identification, if c^I is the CE coordinate dual to the Hamiltonian mode H_I, and u_I is the CE coordinate dual to the shifted cotangent mode η_{δ_I}, then

$$c^I \longmapsto \frac{\partial}{\partial O_I}, \qquad u_I \longmapsto O_I.$$

The map is an isomorphism of completed commutative dg algebras and of P_0-algebras. The induced moment-map shadow is

$$f \longmapsto O_f = \mathrm{Tr}\, f(\phi_1, \phi_2),$$

and satisfies

$$\{O_f, O_g\} = O_{\{f,g\}}.$$

On intervals this construction extends by compactly supported current algebras and gives a morphism of locally constant factorization algebras

$$\mathrm{Obs}^{cl}_{\mathrm{closed},H}{}_{\mathbb{R}\times p} \longrightarrow Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{cl}_{\mathrm{open},H}).$$

The PBW/Koszul-dual open descendant sector is recovered separately as the special fibre of

$$U(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]),$$

and its deformation-level boundary realization uses the principal-part module $P \cong \mathfrak{h}^\vee_{\mathrm{cont}}$, not the polynomial $\Psi_{a,b}$ representatives.

That is the actual open-closed derived center structure.

Sources

Sources

I want it to be a dirac-inspired paper (as if it was written by dirac) on the physics here, that makes natural the factorisation algebra from physical first-principles etc. locality (or lack thereof) is treated physically. unitarity or lack thereof...

Thought for 5m 44s

Yes. The paper should be **Diracian** in the following sense: begin with observables, constraints, brackets, and locality of measurement. Do not begin by asking whether this is "really a string theory." The physics is canonical mechanics with constraints, lifted to BV/factorization language.

The central statement should become:

$$\text{The closed Hamiltonian BF sector is the derived Poisson center of the open brane p}$$

The brane does not merely receive closed Hamiltonians by substitution $z_i \mapsto \phi_i$. That substitution is the **moment-map shadow** of a larger statement: closed local fields act as all central operations, including multiplication, Hamiltonian vector fields, and higher polyvector operations. This makes the factorization algebra natural from first principles.

Your current draft already has the ingredients: the mixed local model, the open ghost-zero action

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1 d\phi_2 + A[\phi_1, \phi_2]\big),$$

the Hamiltonian BF closed sector, and the unresolved full P_0 factorization-center lift in Problem 0.2.20. The rewrite should not present that lift as an external technical problem. It should be the physical heart of the paper. main

Below is the physics-first structure I would use.

1. The Dirac principle of the paper

A local theory is not primarily a Lagrangian. It is a rule assigning observables to regions.

If U is a region, write

$$\mathrm{Obs}(U)$$

for the cochain complex of measurements supported in U. If

$$U_1, \dots, U_n \subset V$$

are disjoint, measurements in the U_i may be performed independently and then regarded as one measurement in V. This gives the structure map

$$\mathrm{Obs}(U_1) \otimes \cdots \otimes \mathrm{Obs}(U_n) \longrightarrow \mathrm{Obs}(V).$$

This is the factorization product. No category theory is being imposed from outside. It is just the algebraic expression of independent local experiments.

If the theory is topological in a direction, moving an insertion in that direction does not change its physical class. Infinitesimally this means

$$L_v = [Q, \iota_v].$$

Thus translations are Q-exact. Factorization is then locally constant in that direction.

If the theory is holomorphic in a complex direction, antiholomorphic translations are Q-exact:

$$L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}].$$

Thus the theory is not local in the Lorentzian sense. It is local in the operational sense: observables have supports, independent supports multiply, and motion in the topological or antiholomorphic directions is homotopically invisible.

This should be the first-principles origin of factorization algebras in the paper.

2. The local physical system

The normal holomorphic surface has coordinates

$$(z_1, z_2), \qquad \omega = dz_1 \wedge dz_2.$$

A closed infinitesimal symmetry of this formal symplectic disk is a Hamiltonian vector field

$$X_f = \{f, -\},$$

where constants are invisible. Hence the closed Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1.$$

The bracket is

$$[f, g] = \{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g \quad \bmod \mathbb{C}.$$

Now place N identical probes at the point. The two normal coordinates become two matrices,

$$z_1 \rightsquigarrow \phi_1, \qquad z_2 \rightsquigarrow \phi_2, \qquad \phi_i \in \mathfrak{gl}_N.$$

The symplectic form becomes

$$\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2).$$

This forces the first-order action

$$S_{\text{open}} = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1 \, d\phi_2 + A[\phi_1, \phi_2]\big).$$

The field A is not an extra dynamical coordinate. It is a Lagrange multiplier. It imposes the Dirac constraint

$$[\phi_1, \phi_2] = 0.$$

Equivalently, the conjugation moment map is

$$\mu_\epsilon = -\,\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

The constraint is first class. Its Hamiltonian flow is simultaneous conjugation:

$$\delta_\epsilon \phi_i = [\epsilon, \phi_i].$$

So the open brane phase space is the constrained quotient

$$\{[\phi_1, \phi_2] = 0\}/GL_N.$$

The BV reduction is simply Dirac's constraint theory written homologically:

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0.$$

That is the physical reason the open trace algebra becomes the algebra of invariant functions on commuting matrix pairs.

3. The basic closed-open measurement

A closed Hamiltonian $f(z_1, z_2)$ is measured by the brane by substituting the brane's matrix-valued normal coordinates:

$$J_N(f) = \operatorname{Tr} f(\phi_1, \phi_2).$$

At finite N one must choose an ordering. In the BV cohomology, adjacent swaps are exact because

$$Q \operatorname{Tr}(\psi vu) = \operatorname{Tr}([\phi_1, \phi_2]vu).$$

Thus the ordering ambiguity disappears after imposing the constraint. In the stable connected limit,

$$J(f) = \operatorname{Tr} f(\phi_1, \phi_2)$$

is a well-defined Hamiltonian observable.

The matrix Poisson bracket gives

$$\{J(f), J(g)\}_{\text{open}} = J(\{f, g\}_{\mathfrak{h}})$$

after removal of the scalar Hamiltonian. This is not a mysterious string duality. It is the statement that the probe brane is a representation of the Hamiltonian algebra of the normal symplectic disk.

For monomials,

$$f = z_1^k z_2^l, \qquad g = z_1^m z_2^n,$$

one obtains

$$\{\operatorname{Tr}(\phi_1^k \phi_2^l), \operatorname{Tr}(\phi_1^m \phi_2^n)\} = (kn - lm) \operatorname{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1}).$$

This is the ordinary canonical bracket of the probe system.

4. Why the factorization center is forced

A closed insertion near the brane should define an operation on open insertions. If the closed insertion is moved past an open insertion, no physical change should occur except for controlled contact terms. Therefore a closed observable does not merely define an element of the open algebra. It defines a **central operation** on the open factorization algebra.

For a locally constant open theory on a line, central operations are Hochschild cochains. Classically, because the open theory has a Poisson bracket, one must retain the Poisson compatibility. Thus the target is not the ordinary center but the derived P_0-center:

$$Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}_{\mathrm{open}}).$$

This is the correct physical object.

The ordinary statement

$$f \mapsto \mathrm{Tr}\, f(\phi_1, \phi_2)$$

only sees multiplication by a function. The derived center also sees vector fields and higher polyvectors. Those extra operations are exactly what the closed CE differential demands.

This is the point to emphasize: the CE obstruction disappears only when closed fields are allowed to act as **operations**, not merely as central functions.

5. The derived center theorem

Let

$$A_\partial = S(\mathfrak{h})$$

be the stable connected degree-zero open Hamiltonian algebra, with generators

$$O_f = J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2).$$

The Poisson bracket is

$$\{O_f, O_g\}_\partial = O_{\{f,g\}}.$$

Choose a basis e_I of $\mathfrak{h}$, with structure constants

$$[e_I, e_J] = C_{IJ}^K e_K.$$

Write

$$O_I = O_{e_I}.$$

Then

$$\{O_I, O_J\} = C_{IJ}^K O_K.$$

So $A_\partial = S(\mathfrak{h})$ is the coordinate algebra of the linear Poisson space $\mathfrak{h}^\vee$. Its Poisson bivector is

$$\pi_\partial = \frac{1}{2} C_{IJ}^K O_K \frac{\partial}{\partial O_I} \wedge \frac{\partial}{\partial O_J}.$$

The derived Poisson center is modeled by the Poisson cochain complex

$$Z_{\mathrm{fact}}^{P_0}(A_\partial) \simeq \mathrm{PV}(A_\partial), \qquad d = [\pi_\partial, -].$$

Explicitly,

$$\mathrm{PV}(A_\partial) = S(O_I) \otimes \Lambda(\theta^I), \qquad \theta^I = \frac{\partial}{\partial O_I}.$$

The differential is

$$d\theta^K = -\frac{1}{2} C_{IJ}^K \theta^I \theta^J,$$

and

$$dO_K = C_{KJ}^L O_L \theta^J.$$

Now the closed Hamiltonian BF algebra is the shifted cotangent Lie algebra

$$\mathfrak{g} = T^*[1]\mathfrak{h} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

Let c^I be the CE coordinate dual to $e_I \in \mathfrak{h}$. Let u_I be the CE coordinate dual to the shifted cotangent generator $e^I[1] \in \mathfrak{h}^\vee[1]$. Then

$$C_{\mathrm{CE}}^\bullet(\mathfrak{g}) = S(u_I) \otimes \Lambda(c^I),$$

with differential

$$d_{\mathrm{CE}} c^K = -\frac{1}{2} C_{IJ}^K c^I c^J,$$

and

$$d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J.$$

Define

$$\Phi : C^\bullet_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]) \longrightarrow \mathrm{PV}(S(\mathfrak{h}))$$

by

$$\Phi(c^I) = \theta^I, \qquad \Phi(u_I) = O_I.$$

Then

$$d\Phi(c^K) = d\theta^K = -\frac{1}{2} C^K_{IJ} \theta^I \theta^J = \Phi(d_{\mathrm{CE}} c^K),$$

and

$$d\Phi(u_K) = dO_K = C^L_{KJ} O_L \theta^J = \Phi(d_{\mathrm{CE}} u_K).$$

Therefore

$$\Phi d_{\mathrm{CE}} = d\Phi.$$

Since both sides are freely generated by corresponding variables, Φ is an isomorphism of dg commutative algebras:

$$C^\bullet_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]) \cong Z^{P_0}_{\mathrm{fact}}(S(\mathfrak{h})).$$

This is the missing closed-open derived center structure.

The physical interpretation is simple:

$$\text{closed Hamiltonian ghost} \longmapsto \text{Hamiltonian vector-field operation on the brane},$$

while

$$\text{closed cotangent coordinate} \longmapsto \text{boundary Hamiltonian function } O_f.$$

Thus the boundary observable $O_f = \mathrm{Tr}\, f(\phi_1, \phi_2)$ is not the image of the CE coordinate dual to $f \in \mathfrak{h}$. It is the image of the coordinate dual to the shifted cotangent generator. That is the correct Dirac/BV interpretation.

6. The factorization version on the brane line

For an interval $I \subset \mathbb{R}$, define the compactly supported Hamiltonian current algebra

$$\mathfrak{h}_I = C_c^\infty(I) \otimes \mathfrak{h}.$$

Its bracket is

$$[a \otimes f, b \otimes g] = ab \otimes \{f, g\}.$$

The open algebra on I is

$$A_\partial(I) = S(\mathfrak{h}_I),$$

with generator

$$O_f(a) = \int_I a(t) \operatorname{Tr} f(\phi_1(t), \phi_2(t))\, dt.$$

The bracket is local:

$$\{O_f(a), O_g(b)\} = O_{\{f,g\}}(ab).$$

This locality is exactly the delta-function locality of the canonical equal-time bracket. The test functions multiply because the Poisson bracket is supported where the two measurements meet.

The closed current Lie algebra is

$$\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathfrak{h}_I^\vee[1].$$

Then

$$C^\bullet_{\mathrm{CE}}(\mathfrak{g}_I) \cong \mathrm{PV}(S(\mathfrak{h}_I)) \cong Z^{P_0}_{\mathrm{fact}}(A_\partial)(I).$$

The maps are compatible with extension by zero. Therefore the interval-wise construction defines a morphism of factorization algebras

$$\mathrm{Obs}^{cl}_{\mathrm{closed},H}{}_{\mathbb{R}\times p} \longrightarrow Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{cl}_{\mathrm{open},H}).$$

This is the physically natural closed-open map.

7. Locality and the places where locality changes meaning

The paper should be explicit that there are three different notions of locality.

First, there is **support locality along the brane**. If two open measurements are supported in disjoint intervals, they multiply independently. Their bracket has a delta function, so it vanishes unless the supports meet. This is ordinary canonical locality in one dimension.

Second, there is **topological locality**. Since translations in the topological directions are Q-exact,

$$L_v = [Q, \iota_v],$$

the precise position of an insertion is not physical. Only the collision pattern of insertions matters. That is why the factorization algebra is locally constant.

Third, there is **formal holomorphic locality**. The theory has been completed at the brane point $p \in \mathbb{C}^2$. It is not a theory of finite propagation through $\mathbb{C}^2$. It is a theory of holomorphic jets at p. Antiholomorphic translations are Q-exact, and holomorphic dependence is formal.

The cotangent/principal-part sector is local in this formal sense. A principal part such as

$$z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$$

is not a smooth function spread through the normal plane. It is a residue functional supported at the brane. Physically it is a transverse delta-current, or derivative of a delta-current, at the defect.

Thus the paper should say:

$$\text{The cotangent boundary sector is distributionally local, not polynomially local.}$$

This is not a defect of the model. It is what one expects for momenta or dual fields attached to a brane.

Trying to realize the coadjoint module by ordinary polynomial descendants is like trying to realize momentum as a function of position. The correct realization is by local duals, principal parts, or polyvector operations.

8. The role of the one-ψ descendants

The polynomial one-ψ descendants

$$\Psi_{a,b}$$

are real and useful, but they should not carry the full physical burden. They are stable primitive Koszul classes in the open BV algebra. They identify a PBW special-fibre copy of a shifted dual vector space. But they are not the deformation-level coadjoint boundary field.

The correct statement is:

$$\Psi_{a,b} \text{ are PBW-special-fibre descendants;} \quad \text{principal parts are the true coadjoint bo}$$

The derived center already knows the coadjoint structure through polyvectors:

$$C^{\bullet}_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^{\vee}[1]) \cong \mathrm{PV}(S(\mathfrak{h})).$$

If the paper also wants a PBW/Koszul-dual boundary realization of $\mathfrak{h}^{\vee}[1]$, define a boundary principal-part sector

$$P = \bigoplus_{a,b\geq 0,\ a+b>0} \mathbb{C}\,\rho_{a,b},$$

where

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

The residue pairing is

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}.$$

Then the Hamiltonian action is

$$f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\}.$$

For monomials,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

This is the true coadjoint action.

On the brane line one should package this as the distributional boundary current sector

$$\Theta_\rho(a), \qquad a \in C_c^\infty(I), \quad \rho \in P,$$

with brackets

$$\{O_f(a), O_g(b)\} = O_{\{f,g\}}(ab),$$

$$\{O_f(a), \Theta_\rho(b)\} = \Theta_{f\cdot\rho}(ab),$$

$$\{\Theta_\rho(a), \Theta_\sigma(b)\} = 0.$$

This gives a factorization object because support is still controlled by the product ab. The non-polynomiality is only transverse to the brane, where the object is a local distribution.

9. Unitarity

The paper should not pretend that this local twisted model is unitary in the ordinary sense.

It is not.

The model is complex, BV, ghost-valued, and topological-holomorphic. It has no Lorentzian time, no positive Hamiltonian, no Hilbert space with positive inner product, and no reflection positivity statement. The variables include ghosts and antifields. A BV complex is not a unitary Hilbert space.

The correct consistency conditions are instead:

$$QI + \frac{1}{2}\{I, I\} = 0$$

classically, and

$$QI + \frac{1}{2}\{I, I\} + \hbar\Delta I = 0$$

quantum mechanically.

In words:

$$\text{Classical consistency is the master equation.}$$

$$\text{Quantum consistency is anomaly cancellation plus factorization/OPE associativity}$$

If a larger untwisted theory exists and is unitary, this local model may be its protected twisted sector. But the local sector itself does not remember ordinary unitarity. It remembers BRST cohomology, Ward identities, and deformation quantization.

The right sentence is:

> We do not impose unitarity on a theory in which time translations have been made cohomologically trivial. The replacement for unitarity in the present local sector is the BV master equation and the associativity of the resulting factorization algebra.

That is the honest physical position.

10. How the paper should sound

Here is an insertable opening in the intended tone.

Local measurements and constraints.
The object of study is a local constrained system. The normal plane to the brane is a formal holomorphic symplectic plane with coordinates z_1, z_2. Its infinitesimal canonical transformations are Hamiltonians modulo constants. A stack of N probes represents the two normal coordinates by matrices ϕ_1, ϕ_2. The symplectic potential is therefore

$$\mathrm{Tr}(\phi_1\, d\phi_2),$$

and invariance under change of basis of the probes imposes the first-class constraint

$$[\phi_1, \phi_2] = 0.$$

The gauge field A is the multiplier for this constraint. Thus the open action is forced:

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1 d\phi_2 + A[\phi_1, \phi_2]\big).$$

The BV field $\psi = A^*$ records the constraint by

$$Q\psi = [\phi_1, \phi_2].$$

This is Dirac reduction written as a differential.

A closed Hamiltonian f is observed by the brane by evaluation on the probe coordinates,

$$f(z_1, z_2) \longmapsto \operatorname{Tr} f(\phi_1, \phi_2).$$

This is not yet the closed-open map. It is its degree-zero moment-map shadow. The full closed-open operation must assign to a closed local observable a central operation on all open local observables. Therefore its natural target is the derived Poisson factorization center.

The center.
Let $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$ be the Hamiltonian Lie algebra of the formal symplectic disk. In the stable connected sector the open Hamiltonian algebra is

$$A_\partial = S(\mathfrak{h}),$$

with Poisson bracket

$$\{O_f, O_g\} = O_{\{f,g\}}.$$

The derived P_0-center of A_∂ is the Poisson cochain complex

$$Z^{P_0}(A_\partial) \simeq (\mathrm{PV}(A_\partial), [\pi_\partial, -]).$$

The closed Hamiltonian BF sector is the shifted cotangent Lie algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

Its local observables are

$$C^\bullet_{\mathrm{CE}}(\mathfrak{g}).$$

There is a canonical isomorphism

$$C^\bullet_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]) \cong \mathrm{PV}(S(\mathfrak{h})).$$

If c^I is the CE coordinate dual to $e_I \in \mathfrak{h}$, and u_I is the CE coordinate dual to the shifted cotangent generator $e^I[1]$, the map is

$$c^I \mapsto \frac{\partial}{\partial O_I}, \qquad u_I \mapsto O_I.$$

Thus a closed Hamiltonian ghost acts as an infinitesimal operation on open observables, while the cotangent coordinate gives the boundary Hamiltonian function. This is the closed-open map.

Locality.
The factorization structure is the algebra of independent local measurements. If measurements are supported in disjoint intervals, they multiply. If the intervals move in the topological direction without collision, the motion is Q-exact and changes nothing in cohomology. If two insertions collide, the equal-time Poisson bracket gives the contact term.

In the holomorphic normal direction the theory is formal. It is a theory of jets at the brane point. Coadjoint modes are principal parts, hence distributions supported at the brane. This is local in the sense appropriate to a defect: the support is on the defect, while the transverse behavior is encoded by residues.

Unitarity.
The model is a twisted BV sector. It is not a Lorentzian unitary theory. Its consistency condition is not positivity of a Hilbert space inner product, but the classical and quantum master equations. If the model is obtained from a unitary parent theory, unitarity belongs to the parent. The local sector retains only the cohomological algebra of protected observables.

11. The actual thesis sentence

The paper's thesis should be something like:

A stack of branes is a system of probes for the formal symplectic normal disk.
Dirac reduction of the probe phase space gives the open trace algebra.
Hamiltonian BF theory is the shifted cotangent theory of canonical transformation
The closed observables are precisely the derived Poisson central operations
on the stable open Hamiltonian brane algebra.

This makes the factorization algebra unavoidable. It is not a technical language added after the physics. It is the local algebra of measurements in a constrained canonical system.

Sources

Sources